



Statistical Inferences by Bootstrapped Fuzzy Distributions

Julien Rosset¹ · Laurent Donzé¹

Received: 31 May 2025 / Accepted: 23 December 2025
© The Author(s), under exclusive licence to Springer Nature Singapore Pte Ltd. 2026

Abstract

Classical statistical methods are based on precise data and well-defined distributional assumptions, but many real-world datasets involve inherent imprecision or vagueness, which are better represented through fuzzy sets. This paper addresses the challenge of performing statistical inference when data are fuzzy. The primary objective is to develop a robust approach for fuzzy hypothesis testing and the construction of fuzzy confidence intervals. We propose a bootstrap-based methodology for these tasks, defining fuzzy hypotheses, computing fuzzy p-values, and introducing decision rules tailored to the fuzzy framework. To illustrate the practical application of this method, we conduct an empirical analysis of depression indices in elderly populations, using data from the easySHARE dataset. By comparing the Swiss population with neighboring countries, we demonstrate the practical utility of fuzzy inference methods. Our findings reveal that fuzzy confidence intervals and p-values may lead to differing conclusions compared to traditional crisp methods, highlighting the importance of selecting an appropriate inference technique that aligns with the underlying uncertainty and decision-making context. This work advances classical statistical inference by extending it to handle fuzzy data in a flexible, principled, and computationally feasible manner.

Keywords Fuzzy confidence interval · Fuzzy hypothesis testing procedure · Fuzzy p-value · Fuzzy logic · Fuzzy methods in data analysis

Introduction

Statistical inference plays a central role in research. Over the decades, statisticians have developed a wide range of tools tailored to specific types of data, contexts, and estimation methods. These concepts and procedures are now well established and taught across many academic disciplines. Most teaching focuses on classical statistical procedures, which rely on carefully chosen estimators whose distributions are either derived analytically or assumed. At its core, a statistical test is presented as a decision rule based on

a chosen significance level to reject or not reject a given hypothesis.

Today, most data analyses—and the associated inferences—are performed on so-called crisp data. However, in many real-world scenarios, data are not crisp but inherently imprecise. In such cases, data should be treated as fuzzy. Fuzziness may arise from imprecise measurements, subjectivity, or human perception, often associated with linguistic descriptions and vague concepts.

The purpose of this work is not only to model fuzziness but to handle statistical inference appropriately when working with fuzzy data. Statistical tests must also become fuzzy since the data under analysis are considered and modelled as fuzzy. This raises the challenge of adequately formulating hypotheses and designing suitable decision rules. One key difficulty lies in defining estimators or test statistics consistent with fuzzy data. Their distributions cannot be assumed to follow classical forms (e.g., normality), as the fuzziness of the data induces fuzziness in the estimators themselves. Hence, a fuzzy distribution must be considered.

As a general solution, we propose to approximate the distribution of test statistics using the bootstrap technique.

J. Rosset and L. Donzé contributed equally to this work.

✉ Julien Rosset
julien.rosset@unifr.ch
Laurent Donzé
laurent.donze@unifr.ch

¹ Group ASAM, Department of Informatics, University of Fribourg, Bd de Pérolles 90, 1700 Fribourg, Switzerland

The bootstrapped distribution can then be used to test fuzzy hypotheses and compute fuzzy p-values.

Building on previous work, this study explains the concept of fuzzy hypotheses and how they can be formally stated. We introduce a fuzzy inference procedure based on fuzzy confidence intervals derived from bootstrapped distributions and describe how fuzzy p-values are computed. The decision rules—whether to reject or not the fuzzy null hypothesis—are based on specific measures, which we define and justify. Fuzzy p-values, although informative, can be challenging to interpret, so we also propose a complementary measure to aid interpretation.

While our general discussion focuses on fuzzy inference about distribution parameters, we illustrate our method through an empirical application: inference about the mean of a distribution. Specifically, we analyse data from the SHARE project, using the easySHARE dataset. Our focus is on comparing the depression index of elderly people in Switzerland with those in neighbouring countries.

Section 2 provides a literature review summarising recent developments in fuzzy statistical inference. Section 3 introduces the concepts of fuzzy inference and fuzzy p-values. Section 4 briefly summarises how bootstrap distributions can be used to construct fuzzy confidence intervals and compute fuzzy p-values. Section 5 presents an empirical application that demonstrates the validity of our approach and discusses its limitations. Finally, Section 6 concludes the paper.

Literature Review

Although the use of p-values in statistical inference is sometimes debated, it remains a standard practice in both classical and fuzzy frameworks. Fuzzy statistical inference does not deviate from this tradition. Analogous to classical statistics with crisp data and known test statistic distributions, several approaches have been proposed to compute fuzzy p-values in the context of fuzzy data.

In the case of fuzzy data but crisp hypotheses, [22] introduced a method for deriving fuzzy p-values by applying a crisp distribution to a fuzzy test statistic. For each left and right endpoint of the α -cuts of the test statistic, a p-value is computed. The collection of these values across all α -cuts yields the fuzzy p-value. In contrast, when hypotheses are fuzzy but the data remain crisp, [16] proposed a fuzzy p-value construction using Zadeh's extension principle. In a

subsequent study, [15] considered the fully fuzzy case—both data and hypotheses are fuzzy. However, despite accounting for fuzzy data, the test statistic's distribution was assumed to be crisp.

A different and noteworthy approach was introduced by [11], who developed fuzzy p-values based on fuzzy confidence intervals. His method determines the intersection points between the fuzzy null hypothesis and the fuzzy confidence interval of the test statistic. However, two limitations should be noted: first, the distribution of the fuzzy test statistic is neglected; second, when the fuzzy confidence interval becomes crisp, the method yields extreme p-values (0 or 1), which contradicts classical statistical intuition and highlights a limitation in the generalisation.

Given the natural connection between fuzzy statistical inference and fuzzy confidence intervals, constructing such intervals remains a critical aspect of methodological development. A foundational contribution in this area is found in [12], which provided the theoretical basis for fuzzy inference procedures and defined fuzzy confidence intervals and hypothesis tests in a rigorous framework. Building on this foundation, several authors proposed alternative methods, including [6–8, 13, 14, 23, 24]. However, these contributions often share a critical limitation: they assume a predefined, crisp distribution for the test statistic. This assumption is problematic both theoretically and empirically.

An alternative strategy is offered by [4], who developed a general approach for constructing fuzzy confidence intervals using likelihood ratios. Their method incorporates a bootstrap technique adapted to fuzzy data to estimate the distribution of the likelihood ratio, thereby avoiding the need for predefined distributional assumptions.

Bootstrap techniques have been widely used to approximate unknown distributions for several decades and are naturally applicable to the fuzzy context. Recent contributions in this area include [3] and [10], who proposed algorithms for generating fuzzy bootstrap samples. These methods respectively address the location and dispersion characteristics of fuzzy data and introduce an epistemic perspective on fuzzy bootstrap sampling.

Finally, fuzzy distributions themselves play a central role in all the aforementioned discussions. Two notable contributions should be highlighted. First, [21] extended the classical quantile function to define a fuzzy quantile function. Second, [1] demonstrated how empirical fuzzy distributions can be derived directly from fuzzy data.

Concepts and Methods of Fuzzy Inferences

We are interested in estimating a parameter θ characterising a distribution F , based on a sample. In this context, we assume that both the data and the hypotheses are fuzzy. It is, therefore, natural to express the null hypothesis, denoted $\tilde{H}$, in linguistic terms. For example, one might say: “The value of the parameter θ is more or less equal to θ_0 ,” or equivalently, “The difference between the parameter θ and θ_0 is more or less equal to 0.” Variants of this statement might include expressions such as “more or less less than” or “more or less greater than.”

These ideas can be formulated more precisely as one- or two-sided fuzzy hypotheses:

$$\tilde{H}_0 : \theta = \tilde{0} \quad \text{versus} \quad \tilde{H}_1 : \theta \neq \tilde{0}, \quad \text{or} \quad (1)$$

$$\tilde{H}_0 : \theta \leq \tilde{0} \quad \text{versus} \quad \tilde{H}_1 : \theta > \tilde{0}, \quad \text{or} \quad (2)$$

$$\tilde{H}_0 : \theta \geq \tilde{0} \quad \text{versus} \quad \tilde{H}_1 : \theta < \tilde{0}. \quad (3)$$

Here, $\tilde{0}$ denotes a fuzzy value centred around zero, representing the hypothesised value of the parameter. Triangular fuzzy numbers are used to model both data and hypotheses. In this representation, the null hypothesis takes the form $\tilde{H}_0 = (H_0^L, H_0, H_0^R)$, and the fuzzy value to be tested is given by $\theta_0 = (\theta_0^L, \theta_0, \theta_0^R)$.

In the following, we propose two approaches for performing fuzzy statistical inference. The first approach is based on constructing confidence intervals for the parameter of interest. The second approach involves computing fuzzy p-values associated with a fuzzy test statistic. Both methods rely on bootstrapped fuzzy distributions.

Let $\tilde{T}$ be a fuzzy estimator of the fuzzy parameter $\tilde{\theta}$, with the estimate denoted by $\tilde{t}$. We denote its α -cuts by $\tilde{t}^\alpha$, or equivalently, in interval form, as $(t^{L,\alpha}, t^{R,\alpha})$. Using a bootstrapping technique (see, for example, [3]), we generate B bootstrap replicates of the fuzzy estimator, denoted $\tilde{t}_1^*, \dots, \tilde{t}_B^*$. This yields an empirical fuzzy distribution $\hat{\tilde{F}}$ for the parameter $\tilde{\theta}$. This “augmented” distribution includes both the original estimate $\tilde{t}$ and the B bootstrap replicates.

Based on these elements, we now detail the methods for constructing fuzzy confidence intervals. A complete description is available in [19].

Bootstrapped Fuzzy Confidence Intervals

[12] are perhaps the first to properly define fuzzy confidence intervals (see also, e.g., [3, 5]). Recall that in a frequentist approach, the bounds of a confidence interval are functions

of the sample. We denote these functions by $\pi_i(\cdot), i = 1, 2$. Furthermore, we denote a random sample drawn from the distribution of interest by $X = (X_1, X_2, \dots, X_n)$, its fuzzy equivalent in an epistemic sense by $\tilde{X} = (\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_n)$, and the fuzzy realisations by $\tilde{x} = (\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n)$.

Definition 1 (Fuzzy Confidence Interval.)

Let $[\pi_1, \pi_2]$ be a symmetrical confidence interval for the parameter θ at significance level γ , such that π_1 and π_2 are functions of the observations, i.e., $\pi_1 := \pi_1(x_1, \dots, x_n)$ and $\pi_2 := \pi_2(x_1, \dots, x_n)$. A fuzzy confidence interval $\tilde{\Pi}$ is a convex and normal fuzzy set, for which the respective left and right α -cuts $[\Pi^{L,\alpha}, \Pi^{R,\alpha}]$ are defined as

$$\begin{aligned} \Pi^{L,\alpha} &= \inf\{a \in \mathbb{R} : \exists x_i \in \tilde{x}_i^\alpha, \forall i = 1, \dots, n, \\ &\text{such that } \pi_1(x_1, \dots, x_n) \leq a\}, \\ \Pi^{R,\alpha} &= \sup\{a \in \mathbb{R} : \exists x_i \in \tilde{x}_i^\alpha, \forall i = 1, \dots, n, \\ &\text{such that } \pi_2(x_1, \dots, x_n) \geq a\}. \end{aligned}$$

The membership function $\mu_{\tilde{\Pi}}(x)$ can be written as

$$\mu_{\tilde{\Pi}}(x) = \sup\{\alpha \cdot I_{[\Pi^{L,\alpha}, \Pi^{R,\alpha}]}(x) : \alpha \in [0, 1]\}.$$

This fuzzy confidence interval is said to belong to the $1 - \gamma$ confidence region, such that for all parameter values θ , we have:

$$\mathbb{P}(\Pi^{L,\alpha} \leq \theta \leq \Pi^{R,\alpha}) \geq 1 - \gamma, \quad \forall \alpha \in [0, 1].$$

Assuming that an estimated empirical fuzzy distribution $\hat{\tilde{F}}$ has been obtained for the parameter or test statistic of interest, we propose two methods to construct fuzzy confidence intervals at level $1 - \gamma$.

First, the left and right parts of a fuzzy confidence interval for each α -cut at level $1 - \gamma$ can be computed as:

$$\hat{\pi}_{\gamma/2}^{L,\alpha} = t^{L,\alpha} - (t_{(B+1)(1-\gamma/2)}^{*,L,\alpha} - t^{L,\alpha}) = 2t^{L,\alpha} - t_{(B+1)(1-\gamma/2)}^{*,L,\alpha} \quad (4)$$

$$\hat{\pi}_{1-\gamma/2}^{R,\alpha} = t^{R,\alpha} - (t_{(B+1)(\gamma/2)}^{*,R,\alpha} - t^{R,\alpha}) = 2t^{R,\alpha} - t_{(B+1)(\gamma/2)}^{*,R,\alpha} \quad (5)$$

where $t_{(B+1)(1-\gamma/2)}^{*,L,\alpha}$ is the $(1 - \gamma/2)$ -quantile and $t_{(B+1)(\gamma/2)}^{*,R,\alpha}$ is the $\gamma/2$ -quantile of the augmented bootstrap distribution. The index $(B + 1)$ denotes the use of the augmented distribution. The bootstrapped fuzzy confidence interval at level $1 - \gamma$ is then given by $\tilde{\Pi}^\alpha = [\Pi^{L,\alpha}, \Pi^{R,\alpha}] = [\hat{\pi}_{\gamma/2}^{L,\alpha}, \hat{\pi}_{1-\gamma/2}^{R,\alpha}]$. One can easily

prove that this construction satisfies the definition of a fuzzy confidence interval (see [19]).

The second method consists of building quantile functions of a fuzzy random variable. Shvedov [21, p. 478] provides a detailed definition. Assuming a fuzzy bootstrapped distribution $\hat{F}$ for the fuzzy parameter estimate $\hat{t}$ has been generated, we denote at a given α -cut: $\tilde{t}^\alpha = (t^{L,\alpha}, t, t^{R,\alpha}) \sim \hat{F}^\alpha = (\hat{F}^{L,\alpha}, \hat{F}, \hat{F}^{R,\alpha})$. The fuzzy p -quantiles of this fuzzy empirical distribution are the quantiles of the left, central, and right parts of $\hat{F}$, i.e.,

$$\tilde{Q}(p, \hat{F}) = (Q(p, \hat{F}^{L,\alpha}), Q(p, \hat{F}), Q(p, \hat{F}^{R,\alpha})), \quad \alpha \in (0, 1],$$

where

$$Q(p, \hat{F}^{L,\alpha}) \leq Q(p, \hat{F}) \leq Q(p, \hat{F}^{R,\alpha}), \quad (6)$$

and $Q(\cdot)$ is the quantile function. Based on these quantities, assuming $p < (1 - p)$, we define the interval $I_p^{U,\alpha} = [Q(p, \hat{F}^{L,\alpha}); Q(1 - p, \hat{F}^{R,\alpha})]$. One can also verify that such an interval is a fuzzy confidence interval for the fuzzy estimator at confidence level $1 - p$.

Statistical Inferences by Bootstrapped Fuzzy Confidence Intervals

In [17], we described thoroughly new decision rules for fuzzy statistical inference based on fuzzy confidence intervals. Let us summarise them in the following.

Let $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ be a real-valued function to test the parameter θ . Given a sample of fuzzy data, we apply this function to the fuzzy vector $\tilde{X}$, obtaining a fuzzy number $\tilde{\phi}$, with α -cuts denoted by $\tilde{\phi}^\alpha$. A complete definition of a fuzzy test statistic is provided in [22]. We begin by recalling the definition of a fuzzy test in the sense of [12]; see also Berkachy [5, p.122].

Definition 2 (Fuzzy Test.)

Consider a fuzzy random vector $\tilde{X}_1, \dots, \tilde{X}_n$ related to the original $X_1, \dots, X_n$ in the space Ω . Let δ be a particular value of the interval $[0, 1]$. A function $\tilde{\phi}$ is called a fuzzy test for the null hypothesis H_0 against the alternative H_1 at significance level δ if

$$\sup_{\alpha \in [0,1]} P_{H_0} \{ \omega_i \in \Omega, i = 1, \dots, n : \tilde{\phi}^\alpha (\tilde{X}_1(\omega_1), \dots, \tilde{X}_n(\omega_n)) \in \{1\} \} \leq \delta \quad (7)$$

where $\tilde{\phi}^\alpha$ is the α -cut of $\tilde{\phi}$ and P_{H_0} means that the probability is evaluated under the condition of the null hypothesis, i.e. $\tilde{\phi}$ is constructed to test H_0 .

We want to test the fuzzy null hypothesis $\tilde{H}_0$ against the alternative one $\tilde{H}_1$. The hypotheses express some conjectures about the parameter θ to be tested. Assume that $\tilde{\Pi}$ is a fuzzy confidence interval for the parameter θ at the confidence level $1 - \gamma$, which could have been obtained according to the abovementioned methods.

We define below two fuzzy decision functions:

- $\tilde{\phi}^{\text{rej}}$: degree of rejecting the null hypothesis $\tilde{H}_0$;
- $\tilde{\phi}^{\text{acc}}$: degree of accepting $\tilde{H}_0$.

Their behavior depends on the relative position of $\tilde{H}_0^\alpha$ and the fuzzy confidence interval $\tilde{\Pi}^\alpha$ at each level α . More formally, let $\tilde{\phi}^{\text{rej}}$ and $\tilde{\phi}^{\text{acc}}$, two fuzzy statistical tests on $\mathbb{F}_c^*(\mathbb{R})$, the class of the non-empty compact, convex and normal fuzzy sets on $\mathbb{R}$, such that

$$\tilde{\phi}^{\text{rej}} : (\mathbb{F}_c^*(\mathbb{R}))^n \rightarrow [0, 1], \quad \tilde{\phi}^{\text{acc}} : (\mathbb{F}_c^*(\mathbb{R}))^n \rightarrow [0, 1].$$

For a given α -cut, we assume the following functions, where $\complement$ designates the complement of a set and by $\neg$ the negation of a proposition:

$$\tilde{\phi}^{\text{rej},\alpha}(\tilde{X}_1, \dots, \tilde{X}_n) = \begin{cases} 0 & \text{if } \tilde{H}_0^\alpha \subseteq \tilde{\Pi}^\alpha \text{ or if } \alpha = 1, \\ 1 & \text{if } \tilde{H}_0^\alpha \subseteq (\tilde{\Pi}^\alpha)^\complement, \\ b_\alpha & \text{if } (\tilde{H}_0^\alpha \cap (\tilde{\Pi}^\alpha)^\complement \neq \emptyset \text{ and } \neg(\tilde{H}_0^\alpha \supset \tilde{\Pi}^\alpha) \text{ and } \alpha < 1), \\ c_\alpha & \text{if } (\tilde{H}_0^\alpha \cap (\tilde{\Pi}^\alpha)^\complement \neq \emptyset \text{ and } \tilde{H}_0^\alpha \supset \tilde{\Pi}^\alpha \text{ and } \alpha < 1), \\ 0 & \text{otherwise,} \end{cases} \quad (8)$$

$$\tilde{\phi}^{\text{acc},\alpha}(\tilde{X}_1, \dots, \tilde{X}_n) = \begin{cases} 0 & \text{if } (\tilde{H}_0^\alpha \subseteq (\tilde{\Pi}^\alpha)^\complement \text{ or } \alpha = 1), \\ 1 & \text{if } \tilde{H}_0^\alpha \subseteq \tilde{\Pi}^\alpha, \\ d_\alpha & \text{if } (\tilde{H}_0^\alpha \cap \tilde{\Pi}^\alpha \neq \emptyset \text{ and } \alpha < 1), \\ 0 & \text{otherwise,} \end{cases} \quad (9)$$

where $b_\alpha = \left(\frac{r_R - r_L}{H_0^{R,\alpha} - H_0^{L,\alpha}}\right)$, $c_\alpha = \left(\frac{r_R - r_L - (\Pi^{R,\alpha} - \Pi^{L,\alpha})}{H_0^{R,\alpha} - H_0^{L,\alpha}}\right)$ and $d_\alpha = \left(\frac{a_R - a_L}{H_0^{R,\alpha} - H_0^{L,\alpha}}\right)$. When $\alpha = 1$, $H_0^{R,\alpha} - H_0^{L,\alpha} = 0$, we cannot evaluate b_α , c_α , d_α , hence we have to define a specific rule. The quantities a_L , a_R , r_L and r_R are given by:

$$\begin{aligned} a_L &= \inf(\tilde{H}_0^\alpha \cap \tilde{\Pi}^\alpha), \\ a_R &= \sup(\tilde{H}_0^\alpha \cap \tilde{\Pi}^\alpha), \\ r_L &= \inf(\tilde{H}_0^\alpha \cap (\tilde{\Pi}^\alpha)^\complement), \\ r_R &= \sup(\tilde{H}_0^\alpha \cap (\tilde{\Pi}^\alpha)^\complement). \end{aligned}$$

[17] proved that $\tilde{\phi}^{\text{rej}}$ and $\tilde{\phi}^{\text{acc}}$ are fuzzy tests and justified the following decision rules:

Definition 3 (Decision rules.)

Let be

$$L^{\text{rej}} = \int_{[0,1]} \tilde{\phi}^{\text{rej},\alpha} d\alpha \quad \text{and} \quad L^{\text{acc}} = \int_{[0,1]} \tilde{\phi}^{\text{acc},\alpha} d\alpha. \quad (10)$$

The quantities L^{rej} and L^{acc} are referred to as the levels of rejection and levels of acceptance (i.e., non-rejection) of the fuzzy null hypothesis $\tilde{H}_0$, respectively.

1. If $L^{\text{rej}} = 1$, the null hypothesis $\tilde{H}_0$ is fully rejected. In this case, $L^{\text{acc}} = 0$;
2. If $L^{\text{rej}} = 0$, the null hypothesis $\tilde{H}_0$ is not rejected at all. In this case, $L^{\text{acc}} = 1$;
3. If $0 < L^{\text{rej}} < 1$, the value of L^{rej} indicates the degree to which $\tilde{H}_0$ is rejected. The larger L^{rej} is, the stronger the evidence against $\tilde{H}_0$;
4. If $0 < L^{\text{acc}} < 1$, the value of L^{acc} indicates the degree to which $\tilde{H}_0$ is not rejected. The larger L^{acc} is, the stronger the support for not rejecting $\tilde{H}_0$;
5. If $L^{\text{rej}} > L^{\text{acc}}$, the decision tends toward rejection of $\tilde{H}_0$;
6. If $L^{\text{rej}} < L^{\text{acc}}$, the decision tends toward non-rejection of $\tilde{H}_0$;
7. If $L^{\text{rej}} = L^{\text{acc}}$, the test result is inconclusive, and no decision can be made.

Statistical Inferences by Bootstrapped Fuzzy p-Values

Let $\tilde{T} = (T^L, T, T^R)$, a function of $\tilde{X}$, be a fuzzy test statistic to test $\tilde{H}_0$ and $\tilde{t} = (t^L, t, t^R)$ its fuzzy realisation. We

supposed that we have computed the empirical fuzzy distribution $\hat{\tilde{F}}$ of $\tilde{T}$ by bootstrapping. The fuzzy observed statistic is $\tilde{t} = (t^L, t, t^R)$. Let B be the number of bootstrapped samples. We denote by $\tilde{t}_1^*, \dots, \tilde{t}_b^*, \dots, \tilde{t}_B^*$ the bootstrapped distribution of the test statistic. The fuzzy empirical bootstrapped mean is thus given by $\tilde{t}^* = (\sum_{b=1}^B \tilde{t}_b^*)/B = (\tilde{t}^{*L}, \tilde{t}^*, \tilde{t}^{*R})$. Furthermore, we define the absolute value of the fuzzy test statistic $\tilde{t} = (t^L, t, t^R)$ as follows:

$$\begin{aligned} |\tilde{t}| &= (t_{abs}^L, t_{abs}, t_{abs}^R) \\ &= (\min(|t^L|, |t|, |t^R|), |t|, \max(|t^L|, |t|, |t^R|)), \end{aligned}$$

Then, the centred bootstrapped distribution is given by $\tilde{t}_1^* - \tilde{t}^*, \dots, \tilde{t}_b^* - \tilde{t}^*, \dots, \tilde{t}_B^* - \tilde{t}^*$. We note this distribution by $\tilde{F}^* = (F^{*,L}, F^*, F^{*,R})$. Illustrative examples of such distribution are given by Figs. 3 and 4.

As developed in [18], an empirical approximation for the fuzzy p-value can be computed as the following. For a two-sided hypotheses test as (1):

$$\begin{aligned} p^L &= 2 \cdot \frac{\#(F^{*,L} \geq t_{abs}^R) + 1}{B + 1}, \\ p &= 2 \cdot \frac{\#(F^* \geq t_{abs}) + 1}{B + 1}, \\ p^R &= \min\left(2 \cdot \frac{\#(F^{*,R} \geq t_{abs}^L) + 1}{B + 1}, 1\right), \end{aligned}$$

where $\#$ gives the number of cases for which the condition in parentheses is true. For a one-sided hypotheses test as (2), the fuzzy p-value $\tilde{p} = (p^L, p, p^R)$ is calculated as:

$$\begin{aligned} p^L &= \frac{\#(F^{*,L} \geq t^R) + 1}{B + 1}, \\ p &= \frac{\#(F^* \geq t) + 1}{B + 1}, \\ p^R &= \min\left(\frac{\#(F^{*,R} \geq t^L) + 1}{B + 1}, 1\right), \end{aligned}$$

and for a one-sided hypotheses test as (3), we have:

$$\begin{aligned} p^L &= \frac{\#(F^{*,R} \leq t^L) + 1}{B + 1}, \\ p &= \frac{\#(F^* \leq t) + 1}{B + 1}, \\ p^R &= \min\left(\frac{\#(F^{*,L} \leq t^R) + 1}{B + 1}, 1\right). \end{aligned}$$

We must underline that we centred the fuzzy bootstrapped distribution (left, centre and right parts) around $\tilde{t}^*$. Another possibility would've been to centre it around $\tilde{t}^*$ by writing it as $\tilde{t}_1^* \ominus \tilde{t}^*, \dots, \tilde{t}_b^* \ominus \tilde{t}^*, \dots, \tilde{t}_B^* \ominus \tilde{t}^*$ where $\ominus$ is the fuzzy

subtraction operator. Nevertheless, we did not choose it for empirical reasons. The algorithm 1 summarises the computation steps to find the bootstrapped fuzzy p-values.

Data: $\tilde{x}_1, \dots, \tilde{x}_n$ a fuzzy realisation of the random sample.

Result: Fuzzy p-values.

begin

 Compute: $(\tilde{t}, |\tilde{t}|)$;

for $b = 1$ **to** B **do**

 Draw sample $\tilde{s}_b = \{\tilde{x}_1^*, \dots, \tilde{x}_n^*\}$;

 Compute: $(\tilde{t}_b^*)$;

end

 Compute: $(\tilde{t}^*, \tilde{F}^*, \tilde{p})$;

return $\tilde{p}$;

end

Algorithm 1 Fuzzy p-values by a bootstrapped fuzzy distribution

We explained in [18] that fuzzy p-values $\tilde{p} = (p^L, p, p^R)$ are not always easy to interpret and could be misleading. We propose to construct specific, meaningful measures to avoid the defuzzification of the fuzzy p-value, which could lose a lot of information coming from the fuzziness.

Definition 4 (Rejection and acceptance levels.)

Let $\tilde{p} = (p^L, p, p^R)$ be a fuzzy triangular p-value with the following fuzzy membership function $\mu_{\tilde{p}}(x)$:

$$\mu_{\tilde{p}}(x) = \begin{cases} \frac{x-p^L}{p-p^L} & \text{if } p^L < x < p, \\ \frac{x-p^R}{p-p^R} & \text{if } p \leq x < p^R, \\ 0 & \text{otherwise,} \end{cases}$$

and let us define:

$$P = \int_0^1 \mu_{\tilde{p}} dx, \quad (11)$$

$$A = \int_0^\gamma \mu_{\tilde{p}} dx. \quad (12)$$

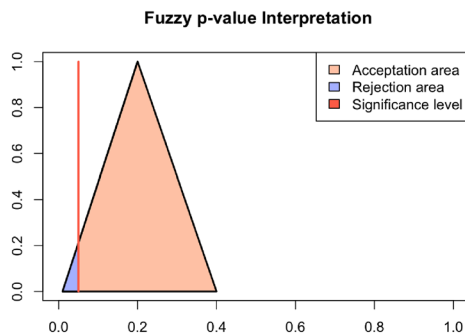


Fig. 1 Example of a fuzzy p-value taken from [18]. Here $\tilde{p} = (p^L, p, p^R) = (0.01, 0.2, 0.4)$. The tendency to reject the null hypothesis (in blue) is $r^{rej} = 0.22$, and the tendency to not reject the null hypothesis (in orange) is $r^{acc} = 0.78$

We call “rejection level” and “acceptance level,” respectively the following two measures:

$$r^{rej} = A/P, \quad (13)$$

$$r^{acc} = 1 - r^{rej}. \quad (14)$$

The real number r^{rej} is between 0 and 1 and tells us how much the fuzzy p-value tends to reject the null hypothesis. On the opposite, $r^{acc} = 1 - r^{rej} \in [0, 1]$ gives us the tendency of not rejecting the null hypothesis. If $r^{rej} > 0.5$, we will tend to reject the null hypothesis, while if $r^{rej} < 0.5$, we will tend not to reject it. No decision can be taken for $r^{rej} = 0.5$. r^{rej} and r^{acc} can be seen as weights to reject or not reject the null hypothesis, respectively. Figure 1 depicts the tendency to reject the null hypothesis in blue. This area equals to $r^{rej} = 0.22$. The orange region has an area of $r^{acc} = 1 - r^{rej} = 0.78$ and gives the tendency not to reject the null hypothesis.

Testing the Mean of the Distribution

As a practical illustration of the proposed methodology, we now consider a fuzzy statistical inference problem for the mean of a distribution. A fuzzy estimator of the population mean is given by the fuzzy sample mean:

$$\tilde{X} = \frac{1}{n} \sum_{i=1}^n \tilde{X}_i$$

with its realisation denoted by

$$\tilde{x} = \frac{1}{n} \sum_{i=1}^n \tilde{x}_i.$$

The bootstrapped distribution of this estimator can be easily generated, and a confidence interval at confidence level γ can readily be computed. Subsequently, the inference procedure described in Subsection 3.2 can be applied. Several illustrative examples are provided in [17].

To implement the second inference approach introduced in Subsection 3.3, namely the use of bootstrapped fuzzy p-values, we consider the following test statistic:

$$\tilde{T} := (\tilde{X} - \theta_0) \oslash \tilde{\sigma}_{\tilde{x}}, \quad (15)$$

where $\oslash$ denotes the fuzzy division operator, and $\tilde{\sigma}_{\tilde{x}}$ is the fuzzy standard error of the sample mean $\tilde{X}$. After generating the bootstrapped distribution of $\tilde{T}$ as described above, the fuzzy p-values can then be computed. A range

Table 1 Index eurod by countries (counts).
Source: easySHARE data, wave 9 ([20])

eurod	0	1	2	3	4	5	6	7	8	9	10	11	12
Switzerland	439	419	363	249	152	106	49	23	12	7	1	1	0
Austria	826	741	570	397	289	229	98	62	31	15	9	1	0
France	408	518	541	418	337	232	169	102	46	23	10	4	1
Germany	754	887	827	714	486	316	171	114	55	31	11	2	0
Italy	843	622	486	414	318	221	177	140	108	84	45	17	5

Table 2 Depression index eurod's fuzzy means per country and their associated fuzzy confidence intervals at confidence level of 95%.

Source: easySHARE data, wave 9 ([20])

Country	Fuzzy mean	Fuzzy confidence interval
CH	(1.254, 2.012, 3.012)	(1.1812, 1.9258, 2.0928, 3.1005)
AU	(1.357, 2.104, 3.104)	(1.2955, 2.0337, 2.1766, 3.1720)
GR	(1.664, 2.491, 3.491)	(1.6069, 2.4331, 2.5534, 3.5462)
FR	(1.930, 2.784, 3.784)	(1.8583, 2.7013, 2.8686, 3.8672)
IT	(1.993, 2.751, 3.749)	(1.9069, 2.6583, 2.8400, 3.8420)

of examples and results are discussed in [18], highlighting the interpretative challenges often encountered with fuzzy p-values, one of them being the tendency of fuzzy p-values to quickly becoming imprecise when fuzziness increases in the observations.

Finally, it is worth noting that the inference decisions derived from confidence intervals and those based on fuzzy p-values may diverge in certain situations (Table 1).

Empirical Application

To illustrate the application of the proposed fuzzy statistical methods, we use data from the [Survey of Health, Ageing and Retirement in Europe \(SHARE\)](#), a large-scale longitudinal study that has been collecting data on Europeans aged 50 and over for nearly two decades. The dataset provides valuable insights into various aspects of aging, including health, living conditions, and socioeconomic status. For the purposes of this study, we focus on the easySHARE dataset [9], which has been specifically designed by the SHARE team for teaching and demonstration.

We analyze the depression index (denoted eurod) recorded in the dataset, which measures depressive symptoms based on twelve items in the mental health module of the SHARE survey. These items assess aspects such as mood, suicidality, appetite changes, and sleep disturbances, with scores ranging from 0 (no depression) to 12 (severe depression). For our analysis, we focus on five countries: Switzerland (CH), Austria (AU), Germany (GR), France (FR), and Italy (IT).

To apply fuzzy methods, we fuzzify the crisp data of eurod using a standard technique for Likert scales, resulting in triangular fuzzy numbers. Specifically, we use the following fuzzification rule:

$$\widetilde{\text{eurod}} = (\max(\text{eurod} - 1, 0), \text{eurod}, \min(\text{eurod} + 1, 12))$$

This moderate fuzzification allows us to explore how fuzzy statistical methods operate on data with inherent imprecision. The next step involves computing the **fuzzy mean** for each country's depression index, followed by bootstrapping to estimate the fuzzy distribution and obtain fuzzy

Fig. 2 Fuzzy confidence intervals at 95% confidence level of the different fuzzy means of eurod per country. The values of these fuzzy confidence intervals are displayed in Table 2

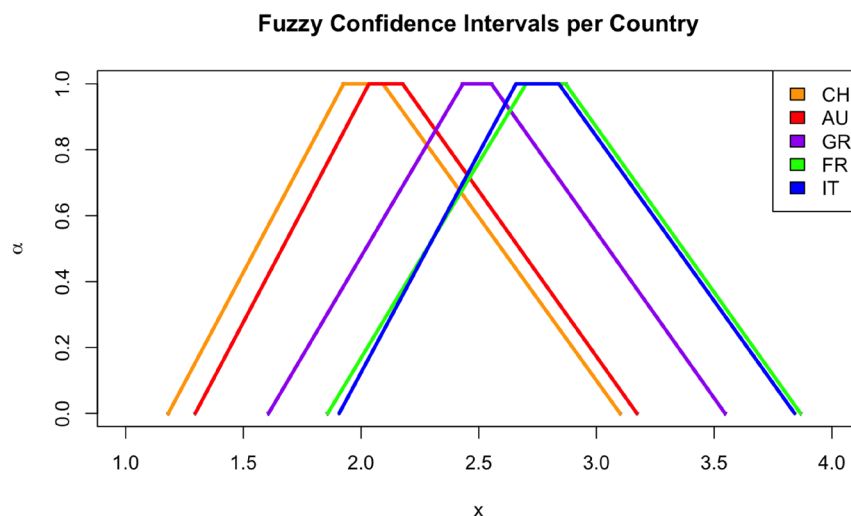


Fig. 3 Fuzzy empirical distribution $\tilde{F}$ of the Swiss's eurod fuzzy mean

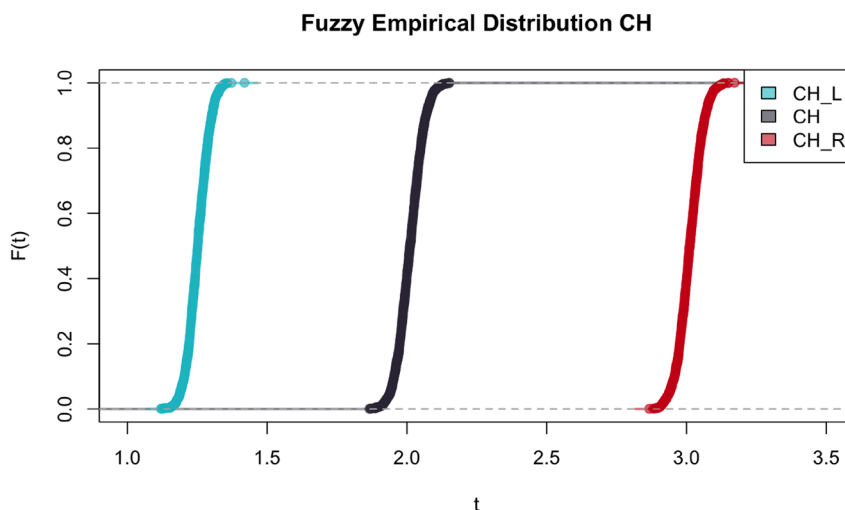
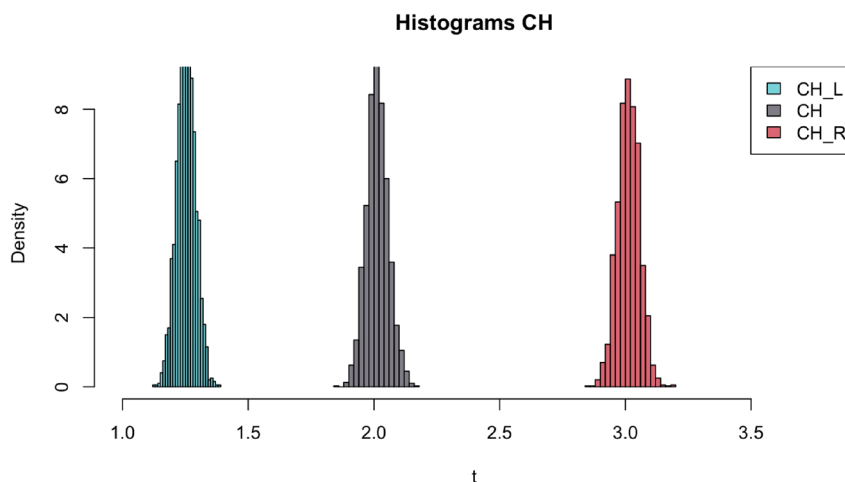


Fig. 4 Histogram of Swiss's eurod fuzzy mean



confidence intervals (FCI). The results of these computations are summarized in Table 2. Figures 2, 3, and 4 illustrate the fuzzy confidence intervals for each country, the Swiss fuzzy distribution, and the Swiss fuzzy histogram of the depression index mean, respectively.

Our primary goal is to compare the fuzzy means of each country's eurod index with the fuzzy mean of Switzerland's eurod index. To clarify, we are not testing for equality of the entire distributions but rather testing if the mean of one country is statistically equal to Switzerland's fuzzy mean.

Table 3 Fuzzy hypotheses testing at a 5% significant level and fuzzy p-values

Tests	$\tilde{H}_0$	L^{acc}	L^{rej}	Fuzzy p-value	r^{acc}	r^{rej}
AU vs CH	(1.254, 2.012, 3.012)	0.964	0.430	(0, 0.009, 1)	0.933	0.067
CH vs AU	(1.357, 2.104, 3.104)	0.990	0.415	(0, 0.042, 1)	0.907	0.093
GR vs CH	(1.254, 2.012, 3.012)	0.615	0.559	(0, 0, 1)	0.9025	0.0975
CH vs GR	(1.664, 2.491, 3.491)	0.615	0.556	(0, 0, 1)	0.9025	0.0975
FR vs CH	(1.254, 2.012, 3.012)	0.411	0.749	(0, 0, 1)	0.9025	0.0975
CH vs FR	(1.930, 2.784, 3.784)	0.397	0.768	(0, 0, 1)	0.9025	0.0975
IT vs CH	(1.254, 2.012, 3.012)	0.396	0.751	(0, 0, 1)	0.9025	0.0975
CH vs IT	(1.993, 2.751, 3.749)	0.395	0.778	(0, 0, 1)	0.9025	0.0975

This table shows the levels of acceptance (L^{acc}) and rejection (L^{rej}) given by the fuzzy testing procedure and the levels of acceptance (r^{acc}) and rejection (r^{rej}) of the fuzzy p-values. The values in Bold contradict the results given by L^{acc} and L^{rej} of the testing procedure

Statistical Testing and Fuzzy p-Values

Table 3 reports the results of fuzzy hypothesis tests at the 5% significance level. The test evaluates whether the fuzzy mean of a country's depression index is approximately equal to Switzerland's observed fuzzy mean. Each pairwise comparison is conducted bidirectionally—first from Switzerland's perspective, then from the other country's perspective—providing a full assessment of the similarity between countries.

For example, the row "AU vs CH" tests whether the fuzzy mean of Austria's depression index is equal to Switzerland's observed fuzzy mean. The null hypothesis $\tilde{H}_0$ is that the fuzzy means of Austria and Switzerland are equal. The second row "CH vs AU" tests the converse, evaluating if Switzerland's fuzzy mean is equal to Austria's observed fuzzy mean. As expected, the results are slightly different due to the directionality of the tests, but both tests yield consistent decisions: they do not reject the null hypothesis, suggesting that the fuzzy means of Switzerland and Austria are approximately equal.

A similar pattern is observed in comparisons with Germany. The fuzzy acceptance level $L^{acc} = 0.615$ and rejection level $L^{rej} \approx 0.556$ suggest a moderate preference for accepting the null hypothesis. This means that, from Switzerland's perspective, there is a 61.5% acceptance that the fuzzy means are equal, and a 55.6% rejection that they are not.

In contrast, comparisons with France and Italy show markedly different results. In both directions, the rejection levels exceed 74%, indicating that Switzerland's fuzzy mean is significantly different from those of France and Italy. This is visually supported by Fig. 2, where Austria and Switzerland exhibit overlapping confidence intervals, Germany shows moderate overlap, while France and Italy have clearly separated intervals.

Fuzzy p-Values and Interpretation

We also report the corresponding **fuzzy p-values** for each comparison in Table 3. Notably, we observe that, for all comparisons the fuzzy p-values are (0, 0, 1), except for the "AU vs CH" comparison, where the p-values are (0, 0.009, 1) and (0, 0.042, 1), indicating near-complete acceptance of the null hypothesis with a level of acceptance $r^{acc} \geq 0.9$ and a level of rejection $r^{rej} \leq 0.1$.

Interestingly, the fuzzy p-values thus indicate a strong tendency to accept the null hypothesis across all comparisons, even in cases where the hypothesis test procedure (based on fuzzy confidence intervals) suggests strong rejection, as, for example, in the tests "FR vs CH" and "IT vs CH". This discrepancy highlights a fundamental issue in fuzzy statistical inference: the two methods (fuzzy confidence intervals and fuzzy p-values) can lead to divergent conclusions. As explained in [18], in cases of highly imprecise data, the fuzziness in the data may cause the fuzzy p-values to offer weaker evidence against the null hypothesis, even when the confidence intervals suggest a clear difference. Due to this limitation, we advocate the use of the fuzzy hypothesis testing procedure which yields much more nuanced results.

Interpretation and Challenges

These contrasting results emphasise the importance of carefully interpreting fuzzy statistical tests and understanding the limitations of different fuzzy inferential methods. The discrepancies between fuzzy p-values and fuzzy confidence intervals underscore the challenge of drawing consistent conclusions from fuzzy data. While fuzzy p-values may seem appealing due to their simplicity and conceptual elegance, they can sometimes be difficult to interpret reliably, particularly when dealing with imprecise or noisy data where fuzzy p-values tend to be close to (0, 0, 1) and thus being strongly in favour of the null hypothesis. In contrast, the fuzzy hypothesis testing procedure seems to be able to

better take into account the fuzziness of the data and provides more refined results.

In other words, as noted in Subsection 3.4 and discussed in [18], fuzzy p-values, despite their potential advantages, may not always provide precise or consistent decisions, especially when the data exhibits significant fuzziness. This reinforces the importance of using the fuzzy hypothesis testing procedure due to its potential to yield more nuanced conclusions than fuzzy p-values.

Conclusion

This study extends classical statistical inference techniques into the fuzzy framework, with a focus on hypothesis testing for fuzzy-valued data. We proposed two complementary methods: (1) the construction of fuzzy confidence intervals using bootstrap resampling, and (2) the calculation of fuzzy p-values derived from bootstrapped fuzzy test statistics. These methods provide new tools for addressing uncertainty and imprecision in statistical inference, an important consideration when working with fuzzy data.

The introduction of fuzziness in data and model parameters introduces significant challenges, both in interpretation and in the decision-making process. Fuzzy confidence intervals offer an intuitive approach to understanding the range of possible values for a parameter, providing a robust framework for inference. On the other hand, fuzzy p-values can sometimes be ambiguous, especially in cases of high imprecision or uncertainty in the data. As demonstrated in our empirical application, the two methods may lead to divergent conclusions, highlighting the need for careful consideration when choosing between them.

Our empirical results underscore this divergence. In some cases, such as the comparison between Austria and Switzerland (AU vs CH), both the fuzzy confidence interval-based decision levels (L^{acc} , L^{rej}) and the fuzzy p-values (r^{acc} , r^{rej}) align, offering consistent conclusions. However, in comparisons involving countries like France and Italy, we observe contrasting results: the confidence interval approach suggests rejecting the null hypothesis, while the fuzzy p-values point to a stronger acceptance of the null hypothesis. These discrepancies reveal the interpretative challenges of fuzzy statistical inference and emphasize the need for context-specific decisions about which inference method to use. The decision-making context, as well as the type and degree of fuzziness in the data, should guide the choice of the most appropriate inferential approach.

Despite these challenges, the bootstrap-based framework we have developed offers a flexible, computationally efficient approach for conducting fuzzy statistical inference. It allows researchers to work with imprecise data without

needing to rely on restrictive assumptions, making it a valuable tool in cases where data uncertainty cannot be ignored.

Future research could further explore the theoretical underpinnings of the proposed procedures, refine their implementation, and extend their applicability to more complex models, such as nonlinear regression models.

Author Contributions All authors contributed equally to the conception, development, and writing of the manuscript. All authors have read and approved the final version of the manuscript.

Funding The authors received no financial support for the research, authorship, or publication of this article.

Data Availability This study used the simplified easySHARE dataset provided by SHARE-ERIC. The dataset is publicly available from the SHARE project under the terms of use outlined by SHARE-ERIC. The specific dataset used is: SHARE-ERIC (2024). SHARE-ERIC: easySHARE, Release Version: 9.0.0. SHARE-ERIC. Dataset. <https://doi.org/10.6103/SHARE.easy.900>

Declarations

Conflict of interest The authors declare no Conflict of interest. They have no relevant financial or non-financial interests to disclose and no affiliations or involvement with any organization or entity with any potential Conflict of interest related to the content of this manuscript.

Ethics Approval Not applicable.

Consent to Participate Not applicable.

Consent for Publication Not applicable.

References

1. Arefi M, Viertl R, Taheri SM. Fuzzy density estimation. *Metrika*. 2012;75(1):5–22. <https://doi.org/10.1007/s00184-010-0311-y>.
2. Berkachy R, Donzé L. Testing fuzzy hypotheses with fuzzy data and defuzzification of the fuzzy p-value by the signed distance method. *Proceedings of the 9th International Joint Conference on Computational Intelligence*. 2017;255–64. <https://doi.org/10.5220/0006500602550264>.
3. Berkachy R, Donzé L. Fuzzy confidence intervals by the likelihood ratio with bootstrapped distribution. *Proceedings of the 12th International Joint Conference on Computational Intelligence*, 2020;231–242. <https://doi.org/10.5220/0010023602310242>
4. Berkachy R, Donzé L. Fuzzy confidence intervals by the likelihood ratio: testing equality of means-application on swiss SILC data. *SN Comput Sci*. 2022;3(5):374. <https://doi.org/10.1007/s42979-022-01257-z>.
5. Berkachy R. The signed distance measure in fuzzy statistical analysis. *theoretical, empirical and programming advances. Fuzzy Manag Methods*. 2021. <https://doi.org/10.1007/978-3-030-76916-1>.
6. Buckley JJ. Fuzzy statistics. *Stud Fuzziness Soft Comput*. 2004. <https://doi.org/10.1007/978-3-540-39919-3>.
7. Couso I, Sánchez L. Inner and outer fuzzy approximations of confidence intervals. *Fuzzy Sets Syst*. 2011;184(1):68–83. <https://doi.org/10.1016/j.fss.2010.11.004>.

8. Chachi J, Taheri SM. Fuzzy confidence intervals for mean of Gaussian fuzzy random variables. *Expert Syst Appl*. 2011;38(5):5240–4. <https://doi.org/10.1016/j.eswa.2010.10.034>.
9. Gruber S, Hunkler C, Stuck, S. Generating easySHARE. Guidelines, structure, content and programming. Working paper series 2014;17-2014 (February). www.share-project.org
10. Grzegorzewski P, Romaniuk M. Epistemic bootstrap for fuzzy data. joint proceedings of the 19th world congress of the international fuzzy systems association (IFSA), the 12th conference of the european society for fuzzy logic and technology (EUSFLAT), and the 11th international summer school on aggregation operators (AGOP), 2021;538–545. <https://doi.org/10.2991/asum.k.210827.071>
11. Hryniewicz O. Statistical properties of the fuzzy p-value. *Int J Approx Reason*. 2018;93:544–60. <https://doi.org/10.1016/j.ijar.2017.12.003>.
12. Kruse R, Meyer KD. Statistics with vague data. 1987. <https://doi.org/10.1007/978-94-009-3943-1>.
13. Kahraman C, Otay I, Öztayşi B. In: Kahraman C, Kabak O, editors. Fuzzy extensions of confidence intervals: estimation for μ , σ^2 , and p . Cham.: Springer; 2016. p. 129–54. https://doi.org/10.1007/978-3-319-39014-7_9.
14. Kahraman C, Öztayşi B, Onar SC. Interval-valued intuitionistic fuzzy confidence intervals. *J Intell Syst*. 2019;28(2):307–19. <http://s://doi.org/10.1515/jisys-2017-0139>.
15. Parchami A. Fuzzy decision in testing hypotheses by fuzzy data: two case studies. *Iran J Fuzzy Syst*. 2020;17(5):127–36.
16. Parchami A, Taheri SM, Mashinchi M. Fuzzy p-value in testing fuzzy hypotheses with crisp data. *Springer Stat Papers*. 2010;51:209–26. <https://doi.org/10.1007/s00362-008-0133-4>.
17. Rosset J, Donzé L. New decision rules for fuzzy statistical inferences. In: Ansari J, Fuchs S, Trutschnig W, Lubiano MA, Gil MÁ, Grzegorzewski P, et al., editors. Combining, modelling and analyzing imprecision, randomness and dependence. Salzburg, Austria: Springer; 2024. p. 405–12. https://doi.org/10.1007/978-3-031-65993-5_50.
18. Rosset J, Donzé L. Approximated fuzzy p-values by bootstrapped fuzzy distributions and fuzzy hypotheses testing. *Proceedings of the 16th International Joint Conference on Computational Intelligence*, 2024;387–395. <https://doi.org/10.5220/0012888300003837>
19. Rosset J, Donzé L. Fuzzy confidence intervals by bootstrapped fuzzy distributions. In: Lesot M-J, Vieira S, Reformat MZ, Carvalho JP, Batista F, Bouchon-Meunier B, et al., editors. Information processing and management of uncertainty in knowledge-based systems. Cham: Springer; 2025. p. 247–53. https://doi.org/10.1007/978-3-031-73997-2_21.
20. SHARE-ERIC: easySHARE, Release Version: 9.0.0. SHARE-ERIC. Dataset. <https://doi.org/10.6103/SHARE.easy.900>
21. Shvedov AS. Quantile function of a fuzzy random variable and an expression for expectations. *Math Notes*. 2016;100(3–4):477–81. <https://doi.org/10.1134/s0001434616090157>.
22. Viertl R. Statistical methods for fuzzy data. 2011. pp. 7–270. <http://s://doi.org/10.1002/9780470974414>.
23. Viertl R, Yeganeh SM. Fuzzy confidence regions. 2016. pp. 119–27. https://doi.org/10.1007/978-3-319-39014-7_8.
24. Wu H-C. Statistical confidence intervals for fuzzy data. *Expert Syst Appl*. 2009;36(2):2670–6. <https://doi.org/10.1016/j.eswa.2008.01.022>.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.